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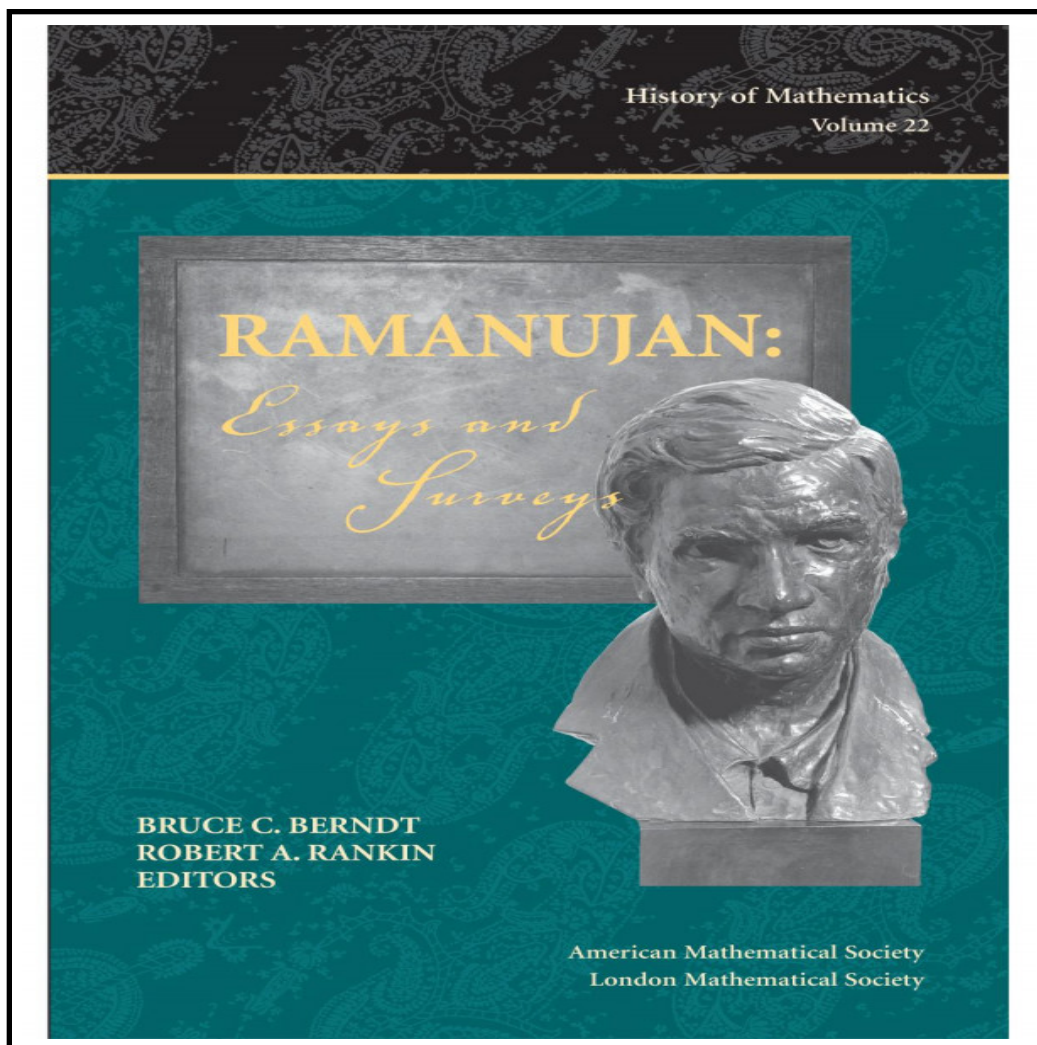
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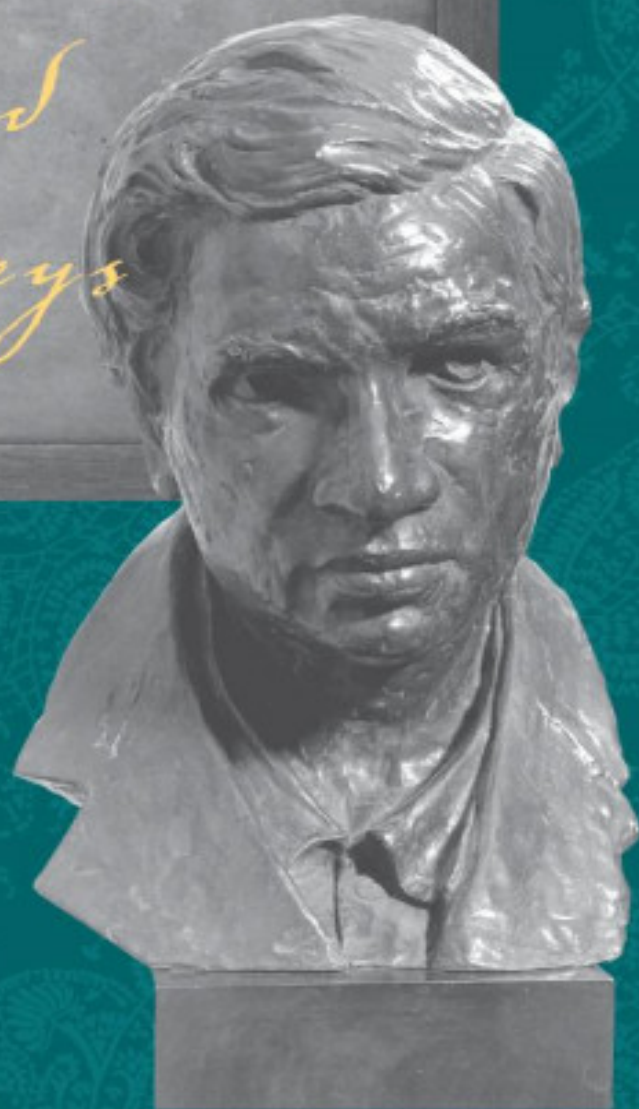
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RAMANUJAN:

Essays and Surveys



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ROBERT A. RANKIN
EDITORS

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Ramanujan Essays And Surveys 1St Edition
Bruce C. Berndt

obituarist Andrew Gray, he published *A Treatise on Bessel Functions and Their Applications in Physics*, which appeared in 1895 and was later translated into Russian.

He resigned his Bangor chair in 1896, being discouraged by the attitude of his students, and returned to a Lectureship in Cambridge. But he returned to Bangor in 1911 to a special Lectureship and resumed some of his professorial duties.

Acknowledgment: We are grateful to J. W. Anderson, T. G. Ramamurthy, K. Srinivasa Rao, Eira J. Scourfield, R. Vengan, and Trevor T. West for the help that they have kindly given.

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THE APPLICATIONS
OF
ELLIPTIC FUNCTIONS

BY
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THE INFLUENCE OF CARR'S SYNOPSIS ON RAMANUJAN

BRUCE C. BERNDT

Almost all biographers of Ramanujan (e.g., P. V. Seshu Aiyar and R. Ramachandra Rao [13, p. xii]) point to G. S. Carr's *A Synopsis of Elementary Results in Pure Mathematics* [10] as the book which kindled the fire of Ramanujan's devotion to mathematics. How much did Carr's *Synopsis* influence Ramanujan? Which published papers and which entries in his notebooks [14] have their seeds in the *Synopsis*? We cannot provide definitive answers to these questions, because we know very little about the other books Ramanujan might have studied in his formative years. However, upon a close examination of Carr's book, we can suggest some topics which Ramanujan *might* have learned from Carr.

At the outset, we emphasize that most of Carr's book seems to have had little influence on Ramanujan's development. The majority of subjects expounded by Carr are not evinced in Ramanujan's writings. In particular, the book contains an enormous amount of geometry, a subject to which Ramanujan evidently devoted little of his creative energy. His published work in geometry encompasses one single-page paper on squaring the circle [11], [13, p. 22], two problems submitted to the *Journal of the Indian Mathematical Society* [13, pp. 331, 333], and a short section in his famous paper on modular equations and approximations to π [12, Sect. 12], [13, pp. 35–36], wherein he used a couple of geometrical constructions to find approximations to π . His notebooks contain six geometrical claims, each accompanied by a figure and all found in Chapters 18 and 19 of his second notebook [14], [3, pp. 190, 194, 211, 243, 245, 299].

Ramanujan possibly first learned differential and integral calculus from the *Synopsis*, for it contains considerable material on these subjects, especially on calculating indefinite integrals. Infinite series are not as thoroughly treated as differential and integral calculus.

In the remainder of this article, we indicate some topics in Carr's *Synopsis* which relate to Ramanujan's papers and notebooks. We begin with elementary mathematics.

Problems submitted by Ramanujan to the *Journal of the Indian Mathematical Society* [13, pp. 327, 329, 334], [9, pp. 18, 20–23] and entries in his notebooks [4, pp. 34–35] indicate Ramanujan's fondness for elementary identities involving radicals. Ramanujan's calculations of class invariants [5, Chap. 34], [7], [8] frequently required elementary radical identities, in particular, denesting theorems. Pages 50–53 in the *Synopsis* treat surds, and, in particular, on page 52 we find the identity

$$\sqrt{a \pm \sqrt{b}} = \sqrt{\frac{1}{2}(a + \sqrt{a^2 - b})} \pm \sqrt{\frac{1}{2}(a - \sqrt{a^2 - b})},$$

which denests if $a^2 - b$ is a perfect square. Undoubtedly, Ramanujan used this identity numerous times to simplify calculations of class invariants.

Certain results in Ramanujan's notebooks require finding the roots of cubic or quartic polynomials. The Cardan and trigonometric methods for solving cubics are found on pages 127–129, and methods for solving quartics are given on pages 129–130 in Carr's text.

Differential calculus begins on page 257, and integral calculus commences on page 313. The attention given to integrals is considerably more extensive. Pages 343–347 are devoted to elliptic integrals with the elliptic integrals of the first and second kinds, $K(k)$ and $E(k)$, respectively, defined on page 344, and the lemniscate integral appearing on page 347. The integral $K(k)$ is prominent in Ramanujan's extensive work on elliptic functions [3], [5, Chaps. 33, 36], and he proved some beautiful theorems on the lemniscate integral [4, Chap. 26]. However, entries in his notebooks [3, pp. 104–113] suggest that Ramanujan was more fully acquainted with elliptic integrals through a text on elliptic functions. In his *Quarterly Reports* [1, pp. 313–318], Ramanujan established a beautiful generalization of the basic theorem on Frullani integrals. Under suitable hypotheses on f , one version of Frullani's theorem asserts that

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = (f(0) - f(\infty)) \log \left(\frac{b}{a} \right).$$

Also a short fragment published with his lost notebook [15, pp. 274–275], [6] features Frullani integrals. The Frullani integral theorem appears on page 392 in Carr's *Synopsis*, with various examples scattered nearby, for example, on pages 380, 381, and 385.

Some elementary theorems on the convergence and divergence of infinite series appear on pages 80–82. Topics in infinite series are interspersed throughout the book. For example, the partial fraction decompositions of trigonometric functions are found on page 426. Ramanujan was especially fond of partial fraction expansions; see several expansions in Chapters 14, 15, and 18 of his second notebook [2, pp. 237, 248–249, 267–275, 277–279, 291–293, 314–315], [3, pp. 200–206] and scattered decompositions in the unorganized pages of his second and third notebooks [4, Chap. 30]. Ordinary hypergeometric series ${}_2F_1$ are defined on page 97, but with the exception of a continued fraction representation for a certain quotient of hypergeometric functions and two corollaries, no further properties are delineated. Ramanujan rediscovered most of the important classical theorems about ordinary and generalized hypergeometric functions, but found many new theorems in this venerable subject as well [2, Chaps. 10 and 11; Chap. 12, pp. 133–145; Chap. 13, pp. 193–205], [5, Chap. 33]. Sections 13–16 in Chapter 3 of Ramanujan's second notebook [1, pp. 69–81] are devoted to expansions derivable from Lagrange's inversion formula, which can be found on page 278 in the *Synopsis*. In fact, one of Ramanujan's examples [1, p. 71], the Maclaurin series

$$\left(\frac{2}{1 + \sqrt{1 - 4t}} \right)^n = 1 + nt + n \sum_{k=2}^{\infty} \frac{\Gamma(n + 2k)t^k}{\Gamma(n + k + 1)k!},$$

is given in an equivalent form by Carr on page 281. Fourier series, which Ramanujan often used, are introduced on page 432. Throughout his notebooks, Ramanujan made numerous applications of the Euler–Maclaurin summation formula; for example see [1, pp. 87, 134–136], [2, pp. 300–302]. Most notably, Ramanujan's theory of

divergent series and his definition of the “constant” of a series (either convergent or divergent) hinges upon the Euler–Maclaurin formula [1, Chap. 6]. The Euler–Maclaurin formula is given in a very imprecise form in terms of indefinite integrals on pages 558–560 in Carr’s *Synopsis*.

It is curious that on page 352 Carr sums two infinite series by setting $n = 1/dx$ in two finite sums of n terms, which can be interpreted as Riemann sums for certain integrals, and then letting dx tend to 0. This peculiar device was also used by Ramanujan in his notebooks [1, p. 27].

Ramanujan’s continued fraction expansions of analytic functions are among his most beautiful contributions to mathematics. The basic theory of continued fractions is sketched on pages 62–66 in the *Synopsis*. Note that the now archaic terms “definite” and “indefinite” continued fractions for convergent and divergent continued fractions, respectively, are defined on page 63; this terminology was used by Ramanujan [5, p. 35]. Although the elementary theory of continued fractions is adequately presented, there is no material on expansions of functions of either a real or complex variable in Carr’s book, except for the aforementioned continued fractions of quotients of hypergeometric functions.

Infinite product representations of a few elementary functions are found on pages 431–432 in Carr’s *Synopsis*. One of Ramanujan’s papers [13, pp. 50–52] and certain entries in his notebooks [1, pp. 38–41], [2, pp. 230–231, 241], [5, p. 560] focus on representing certain functions by infinite products.

Basic properties of the gamma and beta functions are given on pages 359–364, Stirling’s formula for $\log \Gamma(x)$ is on page 412, and considerable material on the logarithmic derivative of $\Gamma(x)$ is located on pages 404–410. The gamma function permeates much of Ramanujan’s work. In particular, basic properties are recorded in Chapter 7 of his second notebook [1, pp. 150–180], analogues of the gamma function are studied in Chapters 8 and 9 [1, pp. 196–231, 273–285], and beautiful continued fractions for quotients of gamma functions are found in Chapter 12 [2, pp. 140–164] and the unorganized portions of his notebooks [5, pp. 50–66]. See also his interesting paper on integrals of gamma functions [13, pp. 216–229].

Ramanujan’s first published paper [13, pp. 1–14] is on properties of the Bernoulli numbers $B_n, n \geq 0$, which are analytically defined on page 276 in the *Synopsis*. Carr offers a recurrence formula for B_n but does not discuss arithmetical properties of B_n , to which Ramanujan devotes Sections 6–10 in his paper. Euler’s formula for $S_{2n} := \zeta(2n) := \sum_{k=1}^{\infty} k^{-2n}$ in terms of B_{2n} , where n is a positive integer, is given on both pages 277 and 430. Formulas for sums related to $\zeta(n)$ are the subject of Ramanujan’s fourth published paper [13, pp. 20–21]. Finally, the dilogarithm $\text{Li}_2(z) := \sum_{k=1}^{\infty} z^k/k^2$ and some of its basic properties are related on pages 368–370. In Chapter 9 of his second notebook, Ramanujan records several basic properties and values of the dilogarithm $\text{Li}_2(z)$, all of which are classical, for example,

$$\text{Li}_2(1-z) + \text{Li}_2\left(1 - \frac{1}{z}\right) = -\frac{1}{2} \log^2 z.$$

However, he also examines some analogues of the dilogarithm, and all of these beautiful results are new [1, pp. 246–260].

It has been suggested by several of Ramanujan’s biographers that Carr’s cryptic style, with lists of theorems followed by sketchy proofs, influenced Ramanujan’s habit of successively recording theorems with no proofs in his notebooks. This may

be true, but there are perhaps two more important reasons for the lack of proofs in his notebooks. First, Ramanujan worked mainly on his slate, for paper was expensive for him. Thus, only the final formulas or theorems were recorded in his notebooks. Second, Ramanujan probably thought that it was unnecessary for him to record any proofs. The notebooks were his own personal compilation of what he had discovered, and if someone had asked him to prove a certain result on a certain page, he knew he could reproduce his argument.

It also might be mentioned that biographers of Ramanujan usually assert that Carr's *Synopsis* contains about 6000 theorems, since the number assigned to the last entry of the book is 6165. However, the numbering contains many gaps, ranging from a gap of one, of which there are several, to a gap of length 170. The present writer has actually counted the number of articles and found the number to be 4417.

Undoubtedly, we have failed to discern some of the threads leading from Carr's *Synopsis* to Ramanujan's papers and notebooks. On the other hand, because so little is known about Ramanujan's early mathematical education, some of the proposed influences of Carr may be illusory. In summary, Carr's idiosyncratic book most likely had a strong influence on Ramanujan's early work, but as the depth of Ramanujan's discoveries deepened, Carr's influence certainly waned. But we can all be thankful for Carr's spark which ignited the flame.

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THE NOTEBOOKS OF SRINIVASA RAMANUJAN

BRUCE C. BERNDT

Ramanujan's three notebooks are kept in the office of the Librarian at the University of Madras. To sustain preservation, all pages have been covered with a thin lamination, apparently applied after the pages were photographed for publication in 1957 by the Tata Institute of Fundamental Research. Unfortunately, this protective coating has not been completely successful, for insects have destroyed small portions of several pages, including the lamination. This damage was evidently absent at the time of photography.

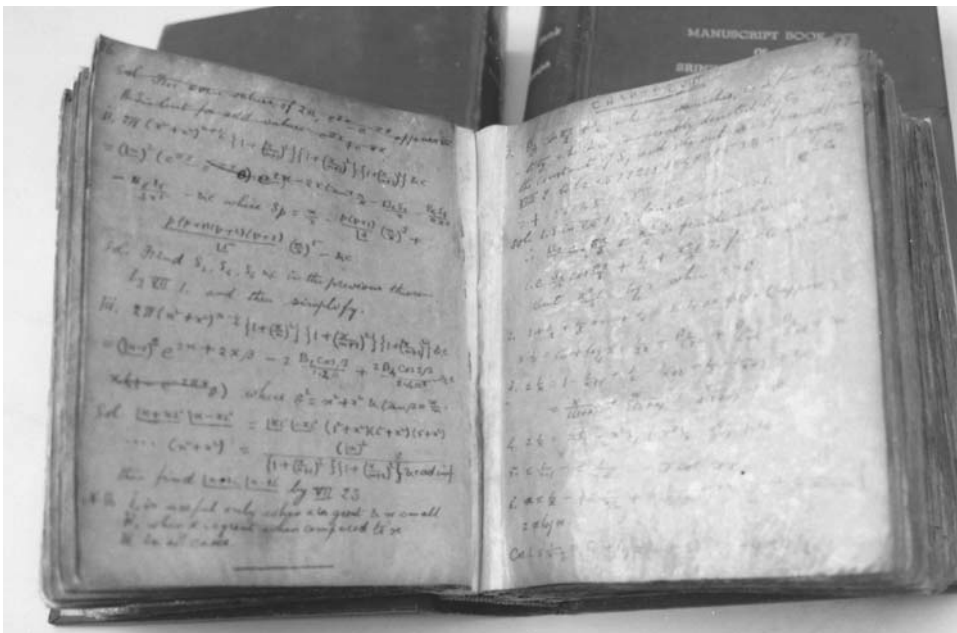
There are some minor differences between the original notebooks and the Tata Institute's edition [1]. Pages 301–356 in [1] are upside down in the original version and proceed from back to front. The first sixteen pages in the third notebook are followed by blank pages comprising a thickness of nearly an inch. The remaining pages are ordered differently in the two versions, with several pages upside down in the original.

The entries are penned in a variety of shades and colors of ink. For example, the first portions of the first notebook are written in green ink. In the first four chapters of the second notebook, theorems are written in black ink with “solutions” and examples prepared in red ink.

A more complete description of Ramanujan's notebooks can be found in Part VI of this book.

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The original notebooks of Ramanujan, now located in the University of Madras Library.

A RECENTLY DISCOVERED LETTER GIVING RAMANUJAN'S EXAMINATION SCORES

University of Madras

The Forge, Kotagiri

My dear Davidson,

Your D O No 1781 W-1 (confidential) of the 4th instant was forwarded to me here.

The Office records show that S. Ramanujan, the Scholar now at Cambridge appeared and failed at the F.A. Examination of 1907, after private study, 4 years after passing the Matriculation Examination of 1903. His record is

Date of birth	1888; (June)
Father's name	Srinivasa Aiyangar, Mirasdar
Matriculated	1903, from Town High Sch. Kumbakonam

His marks in the First Examination in Arts of 1907 were (as Register number 1198)

<u>Subject</u>	<u>Max. Marks</u>	<u>Minimum Required</u>	<u>Marks obtained</u>
English	200	70	38
Sanskrit	100	35	34
Mathematics	150	45	85
Physiology	75	-	-
History	75	-	-

In all probability he absented himself from Physiology and History, which was Roman and Greek.

Commentary. This letter, written on May 15, 1917, was kindly provided by C. A. Reddi, who obtained it from the Tamil Nadu Archives after the publication of the authors' first book, *Ramanujan: Letters and Commentary*. The exam was taken in December 1907. The history of Ramanujan's attempts to pass the F.A. Examination is unclear, because some writers have claimed that he also took the F.A. Examination in December 1906, while others assert that he also took the exam in 1908.

The date of Ramanujan's birth is incorrectly given. A mirasdar is a landlord.

After the letter, in a note dated July 6, 1917, it is remarked by A. G. Cardew that "It certainly seems somewhat of a scandal to the University that a man like this should have failed in the FA Exam."

ON RAMANUJAN¹

S. CHANDRASEKHAR

I cannot clearly say anything that will relate to Ramanujan as a mathematician, particularly in this company which includes, among others, Professors Richard Askey, Bruce Berndt, and George Andrews, who have devoted years to exploring and following his many trails. But I do share with Ramanujan the same cultural background in our early formative years: both of us originate in a common social background – he from Kumbakonam and I from Tanjore, both ancient centers of Tamil culture and not very far apart. Besides, Ramanujan's parents and my own grandparents lived in very similar social and financial circumstances. On this account I can probably visualize Ramanujan's background better than even my younger Indian colleagues of later generations.

With this common background, I can perhaps throw some light on some conflicting statements that have been made about Ramanujan and 'God' by some of his Indian contemporaries. I refer here particularly to the colorful stories concerning Ramanujan's devotion to the Namakkal Goddess.

Quite generally, it may be stated that among those who were brought up in South India during the first two decades of this century, there was (and probably still is) very little correlation between observance and belief. In particular, I can vouch from my own personal experience that some of the 'observances' that one followed were largely for the purposes of not offending the sensibilities of one's parents, relations, and friends.

I can say a good deal on these matters, but I shall only state that I do not accept what has commonly been said and written about Ramanujan's religious beliefs. I corresponded with Hardy on this matter while he was preparing for his Harvard Lectures; and I am personally much more inclined to accept his view as expressed in a letter to me dated February 19, 1936.

. . . And my own view is that, at bottom and to a first approximation, R. was (intellectually) as sound an infidel as Bertrand Russell or Littlewood....

One thing I am sure. R was not in the least the 'inspired idiot' that some people seem to have thought him. On the contrary, he was (except for a period when his mental equilibrium was definitely upset by illness) a very shrewd and sensible person: very individual, of course, and with a reasonable allowance of the minor eccentricities of genius, but fundamentally normal and sane.

And this view of Hardy's is corroborated by K. Ananda Rao, himself a mathematician of distinction, who had been Hardy's student and Ramanujan's contem-

¹Remarks made at the Banquet on June 3, 1987

porary in Cambridge. Ananda Rao is well known and remembered for his contributions to the theory of Tauberian theorems, function-theory and the theory of Dirichlet series. He has written:

In his nature he was simple, entirely free from affectation, with no trace whatever of his being self-conscious of his abilities. He was quite sociable, very polite and considerate to others. He was a man full of humour and a good conversationalist, and it was always interesting to listen to him. On occasions when I met him, we used to talk in homely Tamil. He could talk on many things besides mathematics

This view of Ananda Rao is not surprisingly the same as Hardy's. He has written,

. . . the picture which I want to present to you is that of a man who had his peculiarities like other distinguished men, but a man in whose society one could take pleasure, with whom one could drink tea and discuss politics or mathematics.

Let me now turn to the role of Ramanujan in the development of science in India during the early years of this century.

Perhaps the best way I can give you a feeling for what Ramanujan meant to the young men going to schools and colleges during the period 1915–1930 is to recall for you the way in which I first learned of Ramanujan's name.

It must have been a day in April 1920, when I was not quite ten years old, when my mother told me of an item in the newspaper of the day that a famous Indian mathematician, Ramanujan by name, had died the preceding day; and she told me further that Ramanujan had gone to England some years earlier, had collaborated with some famous English mathematicians, and that he had returned only very recently, and was well known internationally for what he had achieved. Though I had no idea at that time of what kind of a mathematician Ramanujan was, or indeed what scientific achievement meant, I can still recall the gladness I felt at the assurance that one brought up under circumstances similar to my own, could have achieved what I could not grasp. I am sure that others were equally gladdened. I hope that it is not hard for you to imagine what the example of Ramanujan could have provided for young men and women of those times, beginning to look at the world with increasingly different perceptions.

The fact that Ramanujan's early years were spent in a scientifically sterile atmosphere, that his life in India was not without hardships, that under circumstances that appeared to most Indians as nothing short of miraculous, he had gone to Cambridge, supported by eminent mathematicians, and had returned to India with every assurance that he would be considered, in time, as one of the most original mathematicians of the century – these facts were enough – more than enough – for aspiring young Indian students to break their bonds of intellectual confinement and perhaps soar the way that Ramanujan had.

It may be argued, perhaps with some justice, that this was a sentimental attitude: Ramanujan represents so extreme a fluctuation from the norm that his being born an Indian must be considered to a large extent as accidental. But to the Indians of the time, Ramanujan was not unique in the way we think of him today. He was one of others who had, during that same period, achieved, in their judgement, comparably in science and in other areas of human activity. Gandhi, Motilal and

more than usual, he ventured to follow the charitable lady's movements with his eyes, and even to approach her and ask some questions concerning the invalids. The somewhat shy reserve of this friend of the poor inspired in him profound respect, and it seemed to him worthy of all praise and in the best of taste that, instead of boasting of her good works before him, she seemed disturbed and annoyed to have been taken by surprise in the exercise of her functions as a sister of charity.

Gilberte was so afraid of being recognized that she was afraid to let her voice be heard—as if it were not as unfamiliar to the marquis as her face—and waited for the peasant woman to answer his questions. But Jean, fearing that the old woman would fail to play her part intelligently and would betray Gilberte's *incognito* by her awkwardness, kept constantly in front of her and edged her toward the fireplace, glaring savagely at her whenever Monsieur de Boisguilbault's back was turned. Mère Marlot, trembling from head to foot and having no comprehension of what was taking place in her house, did not know which way to turn and prayed fervently that the rain might cease and she be delivered from the presence of these new guests.

At last, somewhat encouraged by the marquis's soft voice and courteous manners, Gilberte made bold to answer him; and as he continued to accuse himself of negligence, she said:

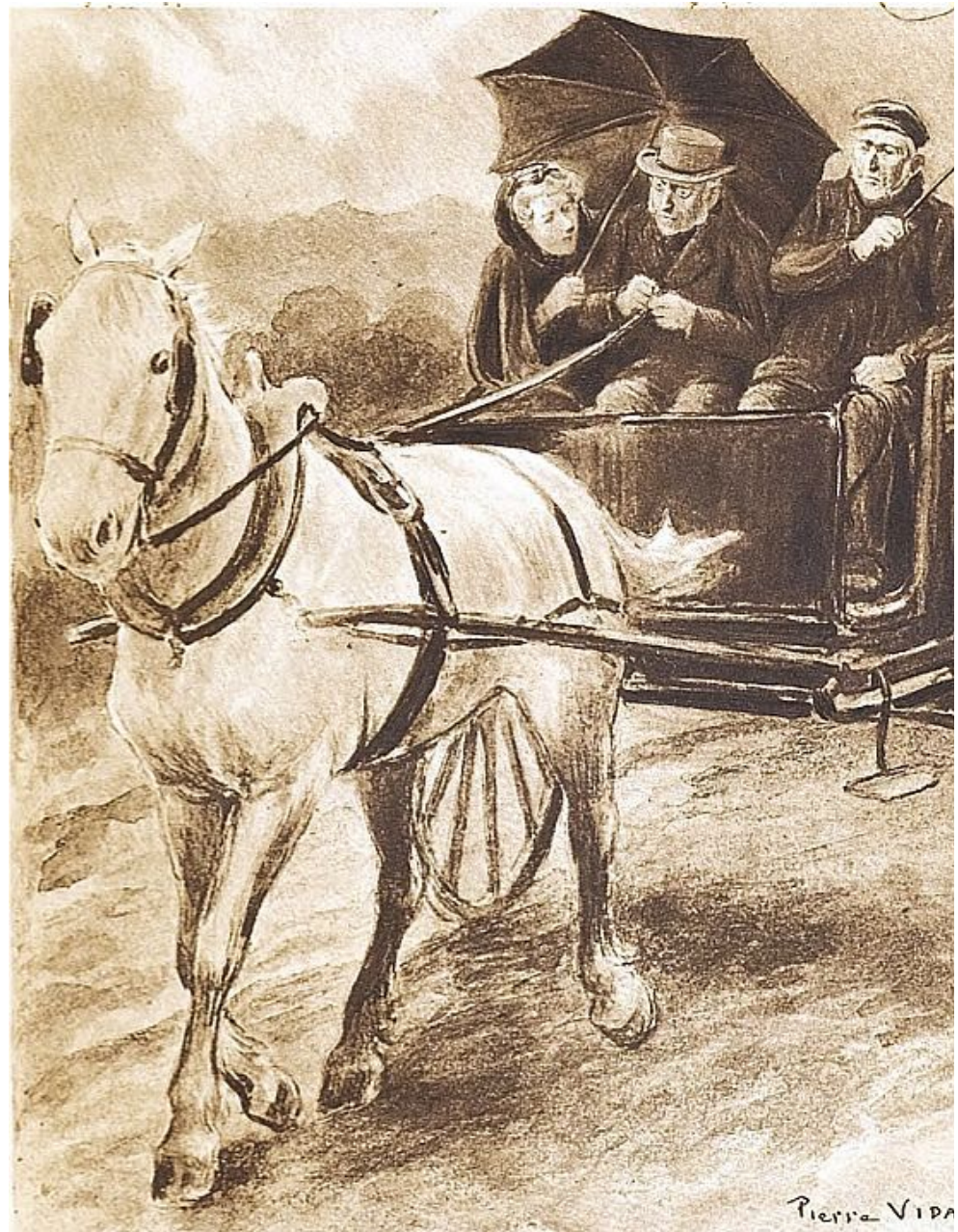
"I have heard, monsieur, that your health is very delicate and that you read a great deal. I can understand that you are unable to attend to so many things as you have on hand. For my part I have nothing better to do, and I live so near that I deserve no great credit for helping to take care of the sick in the parish."

She glanced at the carpenter as she spoke, as if to call his attention to the fact that she was entering into the spirit of her part at last; and Jean hastened to add, in order to give more weight to that pious sentiment:

"Besides, it is a necessity and a duty of her position. If the curé's sister didn't look after the poor, who would?"

"I should be a little reconciled with my conscience," said the marquis, "if madame would kindly apply to me when it happens that I am ignorant or oblivious of my duties. What my zeal leaves undone, my good will can supply; and while madame reserved for herself the noblest and most difficult task, that of nursing the sick with her own hands, I can increase with my money the limited resources of the priest's charity. Allow me to join you in your good deeds, madame, I entreat you, or, if you do not choose to do me

that honor, send all your poor to me. A simple recommendation from you will make them sacred to me."



**GILBERTE AND JAPPELOUP ACCOMPANY
THE MARQUIS TO HIS CHÂTEAU.**

The Marquis took the reins, refusing to allow his charming companion to have the trouble of driving. Jean armed himself with the whip, to stimulate poor Lanterne's courage with a sturdy arm.

"I know that they do not need that, monsieur le marquis," replied Gilberte, "and that you assist many more than I can hope to do."

"You see that is not so, for I have come here entirely by chance, and you are here for the express purpose of doing good."

"Oh no! I did not divine that they needed me," replied Gilberte; "this poor woman came after me; except for that I should probably have known no more about it than you."

"You try in vain to decry your deserts in order to diminish my culpability. They send for you, and they dare not come near me: that fact alone condemns me and glorifies you."

"The deuce! my dear Gilberte," said the carpenter, leading the girl apart, "in my opinion you are performing miracles and you could tame the old owl if you would only have the courage. *Ah but!* as Janille says, all goes well, and if you will act and talk like me, I will answer for it that you will reconcile him with your father."

"Oh! if I only could! but alas! my father has made me promise, yes, swear, that I would never try it."

"And yet he would give all he owns to have you succeed! Look, you, when he made you promise that, he thought that was impossible which is quite possible to-day—not to-morrow perhaps, but this evening, now! We must strike the iron while it's hot, and you can see that there has been a great change already, as he and I came here together and he talks to me in such a friendly way."

"How on earth did that miracle come about?"

"It was a cane that performed the miracle, on my back; I'll tell you about it later. Meanwhile you must be very lady-like, a little bold, and have your wits about you—in a word be like your friend Jean in everything. Listen, I am going to begin!"

Thereupon, Jean abruptly left Gilberte and went to the old man.

"What do you suppose this young lady just whispered in my ear? That she absolutely insists on taking you home in her carriage. Ah! Monsieur de Boisguilbault, you can't refuse a lady; she says that the roads are too badly washed for you to walk, that you are too wet to wait here for your own carriage, that she has a cabriolet with a good horse, a genuine curé's mare that doesn't lose her temper or take fright at anything and goes fast enough when your arm isn't asleep and there's a lash on the whip. In quarter of an

hour you'll be at home, instead of splashing through the mud and stones for an hour."

Monsieur de Boisguilbault thanked the lovely widow warmly but would not accept; but Gilberte herself insisted, with irresistible grace.

"I implore you, monsieur le marquis," she said, turning upon him her beautiful eyes, still frightened like those of a half-tamed dove, "do not pain me by refusing; my carriage is ugly, shabby and muddy, and so is my horse; but they are both strong. I know how to drive and Jean will take me home."

"But it will delay you a long while," said the marquis; "your folks will be anxious."

"No," said Jean, "here is monsieur le curé's page, who serves the mass and rings the bell for him; he's a sure-footed, sharp-eyed rascal, with no more fear of the water than a frog. He has wooden clogs on his feet a little stouter than yours, and he will go to Cuzion as straight and fast as a saw will cut a spruce board. He will tell them not to worry; that madame's in good company and that old Jean will bring her home. So that's settled!—Look you, young wide-awake," he said to Charasson, who yawned as if he would dislocate his jaw and gazed in bewilderment at Monsieur de Boisguilbault; "just come and let me rouse you a bit in the fresh air, and start you on your road."

He dragged, almost carried Sylvain to a short distance from the house, and there, putting his leather apron over his shoulders, he said to him, pulling his ears briskly to fix his words in his memory:

"Run to Châteaubrun and tell Monsieur Antoine that Gilberte is going to Boisguilbault with me; tell him to keep quiet, that all goes well in that direction, and that he needn't worry if she passes the night away from home. Do you hear? do you understand?"

"I hear well enough, but I don't understand," replied Sylvain. "Will you let my ears alone, you old villain of a Jean!"

"I'll make them longer than they are, if you argue; and if you make a botch of my errand, I'll tear them off to-morrow."

"I heard you, that's enough; let me go."

"And if you stop to play on the road, look out!"

"*Pardié!* it's fine weather to play!"

"And if you lose my goatskin apron!"

"I'm no such fool, it won't do me any harm!"

And the child started off at full speed toward the ruins, picking his way in the darkness with the instinct of a cat.

"Now," said Jean leading the old mare and the *barrow* out from under the shed, "it's our turn, honest Lanterne. Oh! don't get excited, Monsieur Sacripant, it's only me! You came with your young mistress, good; but monsieur le marquis, who doesn't look at people, isn't afraid to look at dogs, and he may know you. Do me the favor to follow your friend Charasson. I am sorry to say you must return home on foot."—He cracked the whip at the poor beast and drove him away in the direction Charasson had taken.—"Come, monsieur le marquis, I am waiting for you!" And the marquis, conquered by Gilberte's persistence, mounted the barrow, where he sat between her and Jappeloup.

The stars in heaven did not witness this strange association, for heavy clouds concealed them, and Mère Marlot, the sole witness of this extraordinary adventure, was not sufficiently clear in her mind to indulge in any extended comments. The marquis had put his purse in her hand as he crossed the threshold of her house, and she passed the rest of the night counting the shining coins it contained and waiting on her little ones, saying:

"Dear young lady, she brings us good luck!"

The marquis took the reins, refusing to allow his charming companion to have the trouble of driving. Jean armed himself with the whip, to stimulate poor Lanterne's courage with a sturdy arm. Gilberte, whom Janille, anticipating the storm, had provided with a large umbrella and her father's old cloak when she allowed her to depart on her errand of mercy, gave her attention to sheltering her companions; and as the wind fought for the cloak with her, she held it over Monsieur de Boisguilbault's shoulders with one hand, while she exerted all her strength to hold the umbrella over the old man's head with the other hand, with filial solicitude. The marquis was so touched by these affectionate attentions that he lost all his bashfulness and expressed his gratitude in the warmest terms that his respect would permit. Gilberte trembled at the thought that this sympathetic feeling might change to wrath at any moment, and old Jean laughed in his beard, relying on Providence.

Although it was only nine o'clock, everybody at the château of Boisguilbault had retired when our travellers arrived. No one except old Martin ever paid any attention to the master after sunset, and on this evening

Martin had closed the park after seeing the marquis enter his chalet, and had no suspicion that he had gone abroad and was travelling around the country in the rain and thunder, with an old carpenter and a young woman.

Jean was not particularly anxious to go into the courtyard with Gilberte; for, living so near Châteaubrun as they did, it was impossible that some if not all of the servants should not be familiar with the lovely girl's face, and the first exclamation would betray her.

But the rain was still falling, and there was no plausible excuse for making the marquis or Gilberte alight at the outer gate, especially as Monsieur de Boisguilbault absolutely insisted that his companions should come in and wait by the fire until the rain, which was quite cold and continuous, had ceased. Jean meanwhile was dying with longing to seize this pretext for prolonging the interview; but Gilberte refused in dismay to enter the dreadful manor-house of Boisguilbault, and it was certain that there was great peril in doing it.

Luckily the marquis's eccentric habits made it impossible for them to effect an entrance to the château. In vain did they ring the bell again and again, the wind roared so fiercely that the sound was carried far away. No servant, male or female, slept in that part of the building, where a grewsome solitude habitually prevailed; and, as for old Martin, the only person who ever ventured there, he was too deaf to hear anything, the bell or the thunder.

Monsieur de Boisguilbault was extremely mortified by his inability to show the hospitality which all the circumstances combined to impose upon him as a duty; and he was very angry with himself for having failed to anticipate what had happened. His wrath was on the point of breaking out anew and turning against old Martin, who went to bed with the sun. But at last, suddenly making up his mind what course to pursue, he said:

"I see that I must abandon the idea of getting into my own house, for I shall never make anybody hear unless I send for cannon to take the house by assault; but if madame is not afraid to visit an anchorite's cell, I have another lodging, the key of which never leaves me, where we shall find all that we need to enable us to warm ourselves and rest."

As he spoke he turned the horse's head toward the park, alighted at the gate, opened it himself, and led Lanterne in by the bridle, while Jean squeezed the trembling Gilberte's arm to encourage her to risk the adventure. "God forgive me!" he muttered, "he is taking us to his wooden house, where he passes all his nights evoking the devil! Never fear, Gilberte, I

am with you, and this is the day we are going to turn Satan out-of-doors here!"

Monsieur de Boisguilbault, having closed the gate behind him, bade the carpenter take the reins and follow him at a foot-pace to a sort of gardener's shed where Emile often hitched Corbeau when he came late or expected to stay late; and while Jean busied himself putting poor Lanterne and Monsieur Antoine's barrow under cover, the marquis offered Gilberte his arm, saying: "I am distressed to ask you to walk a few steps on the gravel; but you will not have time to wet your feet, for my hermitage is right here, behind these rocks."

Gilberte shuddered from head to foot as she entered the chalet, alone with that strange old man whom she had always believed to be a little mad, and who now led the way into the darkness. She was somewhat relieved when he opened a second door, and she saw the corridor lighted by a lamp which stood in a niche decorated with flowers. That retreat, so luxurious and comfortable despite its rustic exterior, pleased her exceedingly, and in her youthful imagination, enamored of poetic simplicity, she fancied that she had found the sort of palace of which she had often dreamed.

Since Emile had been admitted to the mysterious chalet, notable improvements had been made there. He had impressed upon the old man that the stoical habits by which he undertook to protest against his own wealth were beginning to be too severe for a man of his years; and, although Monsieur de Boisguilbault was not as yet attacked by any serious infirmity he admitted that he had suffered much from the cold there during the winter. Emile had himself brought from the château carpets, hangings, thick curtains and suitable furniture; he had frequently lighted a fire in the huge stove for protection against the dampness on rainy nights, and the marquis had yielded to the pleasant sensation of being cared for, a sensation entirely mental to him, in which he saw the proof of a zealous and delicate affection. The young man had also rearranged and beautified the room in which he and the old man often took their evening meal. He had made it into a sort of salon, and Gilberte was delighted to place her little feet, for the first time in her life, on superb bearskin rugs, and to gaze in admiration at the beautiful vases of old Sèvres, filled with the rarest flowers, standing on a marble console.

The fireplace, filled with very dry pine cones, blazed up as if by enchantment when the marquis tossed in a piece of burning paper, and the candles, reflected in a mirror, the oaken frame of which was curiously carved

Jawaharlal Nehru, Rabindranath Tagore, J. C. Bose, C. V. Raman, M. N. Saha, S. N. Bose, and a host of others, were in the forefront of the then fermenting Indian scene. The twenties and the thirties were a period when young Indians were inspired for achievement and accomplishment by these men whom they saw among them.

I do not wish to leave the impression that Ramanujan's influence was only in this very generalized sense. I think it is fair to say that almost all the mathematicians who reached distinction during the three or four decades following Ramanujan were directly or indirectly inspired by his example.

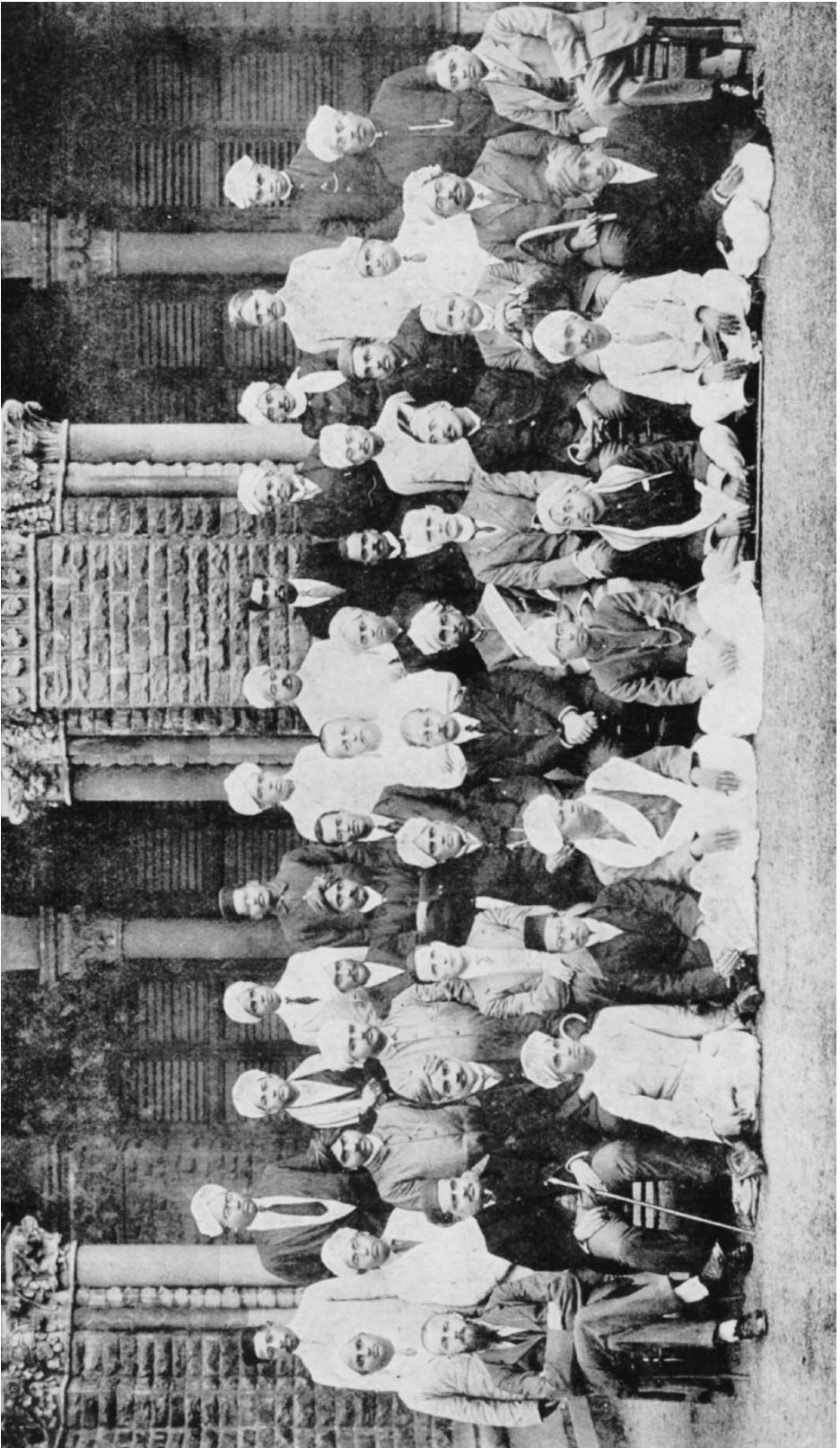
But Ramanujan's name inspired not only ambitious young men planning scientific careers; it also stimulated to action those with public concern. Let me give one example.

When I was a student in Madras one of my classmates (who came from a very wealthy family) was one Alagappa Chettiar. We became good friends; but our lives diverged along different paths after 1930. In the years before and during the second world war, Alagappa Chettiar prospered as an entrepreneur and became a noted philanthropist. He was in fact knighted by the British government.

During the late forties after the war, Sir Alagappa Chettiar (as he was then) wrote to me inquiring if it might be useful for him to found a mathematical institute in Madras named after Ramanujan. I enthusiastically supported the idea; and when I returned to India briefly in 1951, the Ramanujan Institute had been founded a few months earlier. Its first director, T. Vijayaraghavan, was one of the most talented among Hardy's former students; he died at a comparatively early age in 1955. C. T. Rajagopal, a student of Ananda Rao, took over the directorship from him. Already at that time the financial status of the Institute seemed shaky, since Alagappa Chettiar's fortune was melting away.

In April 1957, when Alagappa Chettiar died, the fate of the Institute hung in the balance; Rajagopal wrote to me that the Institute 'will cease to exist on the first of next month,' whereupon I wrote to the Prime Minister (Jawaharlal Nehru), explaining the origin of the Institute and the seriousness of its condition. Nehru's prompt answer was refreshing: 'Even if you had not put in your strong recommendation in favour of the Ramanujan Institute of Mathematics, I would not have liked anything to happen which put an end to it. Now that you have also written to me on this subject, I shall keep in touch with this matter and I think I can assure you that the Institute will be carried on.' And it was; but haltingly and precariously for the next twelve years. It is at this Institute in Madras that Ramanujan's Centennial will be celebrated by an International Conference in December.

There is very little more I can say. My own view, sixty-six years after my first knowing of his name, is that India and the Indian scientific community were exceptionally fortunate in having before them the example of Ramanujan. It is hopeless to try to emulate him. But he was there even as the Everest is there.



THE INDIAN MATHEMATICAL SOCIETY

SECOND CONFERENCE—11TH, 12TH
& 13TH JANUARY 1919—BOMBAY

I. (Standing)—M. L. Chandratreya, C. N. Ganapati, R. Srinivasan, L. S. Vaidyanathan, K. N. Khandekar, S. R. Kashinatha Aiyar, G. G. Pendse, V. M. Gaitonde, K. B. Madhava, V. B. Naik, V. M. Datar, V. Gopalaswami Aiyangar

II. (Standing)—M. Vaidyanathan, S. R. Ranganathan, S. B. Belekar, K. D. Panday, Pandit Hemraj, V. D. Alurkar, C. P. Saldanha, Rev. Steichen, S. Narayanan, R. D. Karve, T. V. Mone, J. C. Swaminarayan, T. K. Venkataraman, M. Bhimasena Rao

III. (Chair)—F. Hallberg, Justice Shah, R. P. Paranjpye, R. Littlehailes, V. Ramaswami Aiyar, A. C. L. Wilkinson, M. T. Naraniengar, Sir John Heaton, D. D. Kapadia, R. Ramachandra Rao, S. V. Ramamurthy, Balakram

IV. (Turf)—C. Krishnamachari, H. R. Kapadia, S. Narayana Aiyar, P. V. Seshu Aiyar, K. S. Patrachari, A. Narasingha Rao, T. M. Patil

This photograph of the participants at the second meeting of the Indian Mathematical Society contains several people important in the life of Ramanujan, including R. Littlehailes, V. Ramaswami Aiyar, R. Ramachandra Rao, S. Narayana Aiyar, and P. V. Seshu Aiyar. S. R. Ranganathan [1] wrote the first book-length biography of Ramanujan. K. B. Madhava, M. Bhimasena Rao, and C. Krishnamachari wrote papers on Ramanujan's work, although it was often not realized that their theorems lay hidden in Ramanujan's notebooks. Finally, R. D. Karve, K. B. Madhava, A. Narasingha Rao, S. Narayanan, M. Bhimasena Rao, J. C. Swaminarayan, and A. C. L. Wilkinson each published solutions to problems submitted by Ramanujan to the *Journal of the Indian Mathematical Society*.

REFERENCES

1. S. R. Ranganathan, *Ramanujan: The Man and the Mathematician*, Asia Publishing House, Bombay, 1967.

